

[Logo: Hackles]

Concepts Lab

Trust your hackles. Pause · Check · Tell.

FIELD COPY

1



Lock It

PASSWORDS AND ACCOUNTS

2



Spot It

SCAMS AND FAKES

3



Share Smart

PRIVACY

4



Safe People

WHO TO TRUST

5



Kind and Well

KINDNESS AND BALANCE

6



Fix It

TELL, REPAIR, RECOVER

PLATE I. The six strands, hand-in-inked, in the order the lessons take them.

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OCTOBER 2026

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Hackles • Cybersecurity Concepts Lab for Teachers

Demo 4: From Secret Codes to Real Encryption

Trust your hackles. Pause • Check • Tell

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Status: DRAFT for teacher, counselor and legal review. Items marked [VERIFY] need checking against a primary source before publication.

AT A GLANCE

Big idea	Encryption scrambles a message so only someone with the key can read it. Toy ciphers fail because there are too few keys and the pattern shows through; modern encryption is designed so that everything except the key can be public.
Grade bands	K-2 (secret picture codes), 3-5 (shift cipher wheel), 6-8 (shift cipher + pattern analysis + padlock box), 9-12 (full, including symmetric vs public-key, key management)
CSTA-style concepts	Networks and the Internet (cybersecurity: encryption), Algorithms and Programming (algorithms, modular arithmetic) (codes: see Concepts Lab Index, Part 4 [VERIFIED 2026-10-05])
Time	15-minute and 45-minute versions
Cost and tech	Paper, scissors, a brad or paper fastener; optional padlocks and a small box. No devices.
Teachers do it first	Yes. Part A is about 20-25 minutes.

MAKE IT STICK

Try This Now (60 seconds, no prep): Write your first name with every letter moved one place forward (A becomes B, Z wraps to A). Swap with a neighbor and decode theirs. How many different shifts would a stranger have to try to crack it? (25.)

01 The big idea

HUMANS HAVE HIDDEN MESSAGES for thousands of years. The core vocabulary has not changed:

- **Plaintext**: the original message.
- **Cipher** (the recipe) and **key** (the secret setting for the recipe).
- **Ciphertext**: the scrambled message.
- **Encrypt** (scramble with the key) and **decrypt** (unscramble with the key).

A classroom shift cipher teaches all of the vocabulary and, importantly, teaches **why it is weak**. Seeing a toy break is what makes it clear why modern encryption is built differently: huge numbers of possible keys, no visible patterns, and a design that stays secure even if the attacker knows exactly how the recipe works.

02 What a teacher should know (plain language)

2.1 THE SHIFT (CAESAR) CIPHER

SLIDE EVERY LETTER FORWARD by the same number of places. With a shift of 3, A becomes D, B becomes E, and so on, wrapping around from Z to A. The **key** is the number (3). Example: MEET AT THE LIBRARY with key 3 becomes PHHW DW WKH OLEUDUB.

Why it is weak:

1. **Only 25 useful keys.** Anyone can try all 25 in a couple of minutes by hand. (This is called a brute-force search.)
2. **Patterns survive.** The most common letter in English text (E) shifts to a different but still most-common letter, so the pattern of letter counts gives it away. This is **frequency analysis**, studied for over a thousand years (it is documented in Arabic scholarship by al-Kindi in the 9th century [historical claim widely reported; the manuscript itself was not checked in Round 3B]).

Understanding this is *defensive* knowledge: it explains why we do not trust home-made codes for anything real.

2.2 KERCKHOFFS'S PRINCIPLE

A cipher should stay secure even if everything about the system except the key is public (stated by Auguste Kerckhoffs in 1883 [historical claim widely reported; "La cryptographie militaire" was not read in Round 3B]). That is why modern encryption standards are published and tested by experts around the world: **secrecy lives in the key, not in a hidden method.**

"Security through obscurity" (hiding the recipe) is not a plan.

2.3 BETTER CIPHERS AND THE PERFECT ONE

- **Substitution cipher** (a scrambled alphabet): about 4×10^{26} possible keys, but frequency analysis still breaks it, because letter patterns still show.
- **Vigenere cipher** (a repeating keyword that shifts each letter differently): MEET AT THE LIBRARY with keyword KEY becomes WICD ER DLC VMZBEPI. It hides single-letter frequencies, but repeating keys can be detected. Historically it was thought unbreakable for centuries until methods were found in the 19th century [historical claim widely reported; not primary-verified].
- **One-time pad**: a random key as long as the message, used **once**. Mathematically unbreakable if truly random and never reused. Impractical for everyday use because it requires delivering as much secret key as there is message. Real-world failures came from reusing pages of keys (the Venona project is the usual example) [historical claim; the NSA Venona page blocked the checker, so it is not primary-verified].

2.4 MODERN ENCRYPTION IN FOUR SENTENCES

1. **Symmetric encryption** (for example AES) uses the *same* secret key to lock and unlock. It is fast. AES keys are 128, 192 or 256 bits; a 128-bit key has about 3.4×10^{38} possibilities, so trying every key is not feasible with any conceivable computer [VERIFIED 2026-10-05: NIST FIPS 197 (updated 2023) specifies AES-128, AES-192 and AES-256 with 128-bit blocks; $2^{128} = 3.4 \times 10^{38}$ recomputed]. Instead of one simple shift, AES scrambles and mixes the data through 10 to 14 rounds of operations (depending on key size) and no practical shortcut is known.
2. **Public-key (asymmetric) encryption** solves the biggest old problem: how do two people who have never met share a secret key? Each person has a **public key** (anyone may use it to lock a message to them) and a **private key** (only they can use it to unlock). It is built on math problems that are easy one way and extremely hard to undo, such as multiplying large primes versus factoring the product (RSA) or related elliptic-curve problems [standard description].
3. Real systems **combine both**: public-key methods are used briefly to agree on a fresh secret key, then fast symmetric encryption protects the actual data. That is what happens when your browser connects to a website (Demo 5).
4. Modern attackers almost never "break the math." They steal the key, trick the user, break the device, or exploit a bug in the software (Demos 3, 7, 11). So encryption protects only as far as **key management** and the rest of the system go.

2.5 THE QUANTUM FOOTNOTE (HONEST AND SHORT)

A sufficiently large quantum computer could break much of today's public-key cryptography (RSA and elliptic curves). Such machines are not believed to exist at that scale today [expert consensus not re-checked in Round 3B; NIST's published post-quantum standards are the evidence that planning has begun]. Standards bodies have already chosen "post-quantum" replacements: NIST published its first three post-quantum standards on August 13, 2024 (FIPS 203, 204, 205) [VERIFIED 2026-10-05: nist.gov news release, Aug 2024]. For teachers: it is a reason to keep software updated (Demo 7), not a reason for alarm.

2.6 COMMON MISCONCEPTIONS

MYTH	BETTER UNDERSTANDING
"Encryption is unbreakable."	Well-designed encryption is extremely strong, but people, keys and software are the weak points.
"Hiding how my code works makes it safer."	No. Real security assumes the attacker knows the method.
"Encryption and hashing are the same."	Encryption is reversible with a key; hashing is one-way (Demo 2).
"Encrypted means private from everyone."	Data may be decrypted by the app or company that holds the keys. End-to-end encryption keeps the keys only with the two people chatting. [standard definition]
"Longer keys always mean better security."	Within a good algorithm, yes up to a point; but a bad algorithm or reused key can ruin it.
"Home-made codes are fine for real secrets."	No. Use well-tested public standards, never invent your own for real use.

03 Materials (zero to low cost)

- **Cipher wheel:** two paper circles (outer 4 inches, inner 3 inches) with 26 evenly spaced letters, joined with a brad; or a paper strip with the alphabet written twice, slid along.
- Padlocks (3-5 cheap ones) and a small lockable box, or draw locks on cards. Index cards, pencils, a timer, the student handout.
- Optional: paint or colored water cups for the color-mixing story (see Script).

04 Part A: Teachers do it first (about 25 minutes)

1. **Build and use a wheel (5 min).** Encrypt "TRUST YOUR HACKLES" with key 7 and check it on the answer page.
2. **Crack it by brute force (5 minutes).** Another teacher gives you a message without the key: GWNLS YMJ INHJ. Try all 25 keys on the wheel. Time yourself and count the tries. (Answer on the teacher page.)
3. **Spot the pattern (5 minutes).** Use the long ciphertext in Section 5 (frequency sample). Tally letters. The most common letter is probably E in the original. See how fast that narrows the key to one.
4. **Padlock box (7 minutes).** Do the Padlock Box activity below. Notice that the receiver's open padlock can be public and nothing bad happens, but the receiver's key must stay private. That is the public-key idea in your hands.
5. **Reflect (3 minutes).** Why can someone know our entire recipe and still not read the message? What goes in the key?

05 Part B: Teaching it

ACTIVITY SHEETS

Cipher Challenge Cards and the frequency sample are printed on the student handout without keys. The keys and plaintexts are on the teacher answer page.

Padlock Box activity (grades 6-12; adaptable to 3-5)

1. Each "receiver" student hands out *open* padlocks (the public key) and keeps the key (the private key).
2. A "sender" writes a message on a card, puts it in the box, snaps on the receiver's open padlock, and passes the box through the whole class (the "network").
3. Everyone in the middle (the "eavesdroppers") can hold the box, shake it, copy the padlock design, but cannot open it.
4. The receiver opens it with the key.
5. Debrief: What was public? (the padlock) What was private? (the key). What if the receiver loses the key? What if someone gave you a fake padlock that *they* hold the key to? (That is why "who really owns this public key" matters; the browser lock/certificate system in Demo 5 is how this is handled.)

Zero-cost version: use envelopes with a stamp "Only Sam can open", and ask students to imagine the physical lock.

The paint-mixing story (optional teacher demo for 6-12; the idea behind Diffie-Hellman key agreement)

- Both people start with the same public color (yellow).
- Each secretly picks a private color (Alice picks red, Bob picks blue), mixes it with yellow, and sends the mixture across the room.
- Each then adds their private color to the mixture they *received*. Both end up with the same final color (yellow + red + blue), but a watcher who saw only the mixtures cannot "un-mix" to find it.
- Say honestly: paint mixing is an analogy; the real math uses one-way arithmetic, not paint. [CS Unplugged is CC BY-SA 4.0 (VERIFIED 2026-10-05: csunplugged.org); check its cryptographic protocols activity and license if you adapt it directly]

15-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-2	Hook: read the secret message "UIF EPH CBSLT" aloud and ask who can read it.
2-5	Show the wheel, vocabulary (key, cipher, plaintext, ciphertext). Encrypt a message together.
5-10	Partners crack Cards 1-3 by trying keys; count tries. Race.
10-13	Discuss: Why was it easy? 25 keys. Show Frequency idea (E is common).
13-15	Bridge to real encryption: more keys, no pattern, key is the secret. Exit slip.

45-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-4	Hook and vocabulary.
4-14	Build wheels; encrypt a message; exchange with a partner using a secret key.
14-22	Crack the Mystery card by brute force; record the number of tries; then crack the Frequency sample using tallies.
22-32	Padlock Box activity and debrief (public vs private).
32-40	"Real encryption" mini-lecture: AES with a table of key sizes; combining public-key and symmetric (browser example); end-to-end encryption; why attackers steal keys instead of breaking math.
40-45	Exit ticket and handout.

06 Teacher script

Hook (2 minutes). "I'm going to write something on the board and I want to see who can read it: UIF EPH CBSLT. No? It's a secret code. By the end of class you will make one, break one, and understand why the codes protecting your messages online are built so differently. Spies and kings have used codes like this. But here's the twist: the code is not secret. The recipe can be on the poster. The **key** is the only secret."

After brute force. "How many tries did it take? Twenty-five at most. That's the problem: too few keys. A real key has more possibilities than a 1 followed by 38 zeros. Even if every computer on Earth tried keys for the age of the universe, they wouldn't finish. [Use the safe version: "far more than any attacker could ever try."]"

After padlock. "We just shared a lock in public and kept the key private. That is the big invention of the 1970s [Diffie and Hellman published "New Directions in Cryptography" in November 1976, VERIFIED 2026-10-05: IEEE Trans. Information Theory IT-22(6), Stanford copy; RSA followed in 1977, not re-checked]: you don't need to meet in secret first."

Close. "People who protect real secrets don't invent their own codes. They use standards that thousands of experts tried to break. Your job as a defender: use good tools (a trusted messaging app, a device lock), keep software updated, and guard your keys, which means your passwords and devices. If someone ever asks you to hand over a key or a code, pause. Check. Tell."

07 Kid-friendly analogies by band

BAND	ANALOGY AND ACTIVITY
K-2	Picture code. Swap each of 6 pictures with a matching picture (star = apple). Kids decode a "mystery message" of pictures. Key = "the matching chart". It is a class game: no secret messages that a grown-up cannot see.
3-5	Cipher wheel. Slide the inner wheel; spies' decoder rings. Keep the key secret, share the method. Crack with all 25 tries.
6-8	Patterns betray you. Frequency counting, the padlock box, the idea that "more keys and no pattern" makes real encryption strong.
9-12	Symmetric vs public-key, key agreement, key management. Why the browser uses both, end-to-end encryption, the limits (metadata, endpoints), quantum caveat, password-derived keys (Demo 1 and 2).

08 Discussion questions

1. Why does a code that only has 25 keys fail? How many keys are enough?
2. If an enemy knows our method but not our key, are we safe? Why?
3. Why did your most common letter give the key away?
4. (9-12) Why is "the math is unbreakable" not the same as "the system is secure"?

09 Level-up challenge

- **Vigenere in a day.** Encrypt with a keyword (KEY) and observe that the same letter encrypts differently each time, which hides frequency. (Stay at the "why more complex keys help" level; do not teach breaking methods.)
- **Key-space calculator.** Compare 26^n for $n = 1$ to 10 with 2^{128} . (26^{10} is about 1.4×10^{14} ; 2^{128} is about 3.4×10^{38} , roughly 2.4×10^{24} times bigger.)

10 Safety and ethics notes (defense only)

- This is a puzzle and history lesson. Do **not** frame breaking messages as hacking real systems; do not suggest breaking into anyone's messages or accounts.
- Real secrets (passwords, personal information) are **never** protected with a toy cipher. Say this clearly. Students must not use the class cipher to hide things from trusted adults: a cipher club game is fine; a secret code used to hide unsafe contact from parents is not. Link to the "No secrets that make you feel uneasy" rule: **tell a trusted adult every time.**
- Do not ask students to attempt to decrypt any real encrypted files or traffic.
- Do not promote home-made ciphers for real use.

11 Assessment idea

- **Exit ticket.** "Name two reasons the shift cipher is weak and one reason modern encryption is stronger." Rubric: 3 = few keys, patterns, huge key space/no pattern; 2 = two of three; 1 = "computers are faster".
- **Performance task (6-12).** Give a new ciphertext (key unknown) and require an explanation of how the key was found, in words or table, not just the answer.

12 Sources and [VERIFY] list

- Auguste Kerckhoffs, "La cryptographie militaire" (1883) [not re-checked].
- Al-Kindi, manuscript on deciphering cryptographic messages (c. 9th century) [not re-checked].
- NIST FIPS 197 (AES), FIPS 203, 204, 205 (post-quantum standards, Aug 2024) [VERIFIED 2026-10-05 for FIPS 203-205; FIPS 197 not checked].
- Diffie and Hellman (1976) "New Directions in Cryptography" [VERIFIED 2026-10-05]; Rivest, Shamir and Adleman (1977) [not re-checked].
- CS Unplugged (University of Canterbury) cryptographic protocols and public key encryption activities, which inspired the padlock and color-mixing ideas (CS Unplugged, University of Canterbury, CC BY-SA 4.0: VERIFIED 2026-10-05). This demo's specific activity text is original; check the license if directly adapting.
- Venona project and one-time pad reuse [not primary-verified].

Student Handout: Secret Codes and Real Encryption

Name: _____

Words to know

- **Plaintext:** the message before it is scrambled.
- **Cipher:** the scrambling recipe.
- **Key:** the secret setting.
- **Ciphertext:** the scrambled message.

The shift cipher

SLIDE EACH LETTER FORWARD by the key. With key 3: A becomes D, B becomes E, Z wraps to C.

PLAINTEXT	KEY	CIPHERTEXT
HELLO	3	KHOOR
_____	5	_____

Cipher Challenge Cards

TRY ALL 25 KEYS on your wheel. Write the plaintext and the key you found.

CARD	CIPHERTEXT	PLAINTEXT	KEY
1	PHHW DW WKH OLEUDUB		
2	XLI GPYF QIIXW EX RSSR		
3	GWNSL YMJ INHJ		
4	ALFDP NSPNV EPWW		
Mystery	UIF EPH CBSLT		

How many tries did Card 3 take you? ____

Frequency challenge (grades 6-12)

THIS MESSAGE USES ONE key. Tally the letters, then guess which letter stands for E.

CQN KNBC FJH CX TNNY J BNLANC RB CX BQJAN CQN TNH XWUH FRCQ
CQN YNXYUN FQX WNNM RC JWM CX TNNY RC BJON

Most common letter: ____ I think it stands for: ____ The key is: ____

Why was it so easy?

1. Only ____ possible keys.
2. The most common letter in English is ____ and it still shows through.

Padlock Box

IN OUR PADLOCK GAME, the **public** part was the _____ and the **private** part was the _____.

Real encryption

- Uses keys with more possibilities than a computer could ever try.
- Hides patterns.
- Stays safe even if the recipe is public. Only the **key** is secret.
- The weak spot is usually people, not math: guard your keys (passwords, devices, recovery codes).

Remember

- Never use a toy cipher for real secrets.
- Do not use a secret code to hide something that makes you feel uneasy from a trusted adult. If something feels off, **Pause. Check. Tell.**

End of student handout.

TEACHER COPY ONLY:

Answers for Demo 4 (do not distribute to students)

FIELD NOTE

No. 1

WARNING: Teacher page. Remove before copying. The student handout ends on the page before this one. Print the handout pages only, or detach this page first. Do not post it, project it or send it home.

Cipher Challenge Cards (verified by recomputing):

CARD	CIPHERTEXT	KEY	PLAINTEXT
1	PHHW DW WKH OLEUDUB	3	MEET AT THE LIBRARY
2	XLI GPYF QIIXW EX RSSR	4	THE CLUB MEETS AT NOON
3	GWNSL YMJ INHJ	5	BRING THE DICE
4	ALFDP NSPNV EPWW	11	PAUSE CHECK TELL
Mystery	UIF EPH CBSLT	1	THE DOG BARKS

Part A check: TRUST YOUR HACKLES with key 7 = AYBZA FVBY OHJRSLZ.

Handout table: HELLO with key 3 = KHOOR; with key 5, HELLO = MJQQT.

Frequency sample (key 9): the plaintext is THE BEST WAY TO KEEP A SECRET IS TO SHARE THE KEY ONLY WITH THE PEOPLE WHO NEED IT AND TO KEEP IT SAFE. Most common ciphertext letters: N (17 times) and C (11). N stands for E (E shifted 9 places forward is N), so the key is 9. In the plaintext, E and T are the most common letters.

Handout answers: "Why was it so easy?" 1: 25 possible keys. 2: E. Padlock box: the open padlock is public; the key is private. Card 3 takes at most 25 tries (the key is 5).

Vigenere example (for the level-up): MEET AT THE LIBRARY with keyword KEY = WICD ER DLC VMZBEPI. Note the same plaintext letter E becomes I in one place and C in others, so the single most-common-letter trick no longer works directly.